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Regular Article

# From feedback to Insight: Uncovering gender bias in STEM evaluations through NLP☆

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ABSTRACT

Gender bias in faculty evaluations threatens equity in higher education. Our study uses Natural Language Processing (NLP) to analyze 70,000 student evaluations from a Latin American STEM university, revealing statistically significant disparities in how students assess male and female professors and exposing implicit bias in qualitative feedback. To achieve this, we developed a framework that combines sentiment analysis, thematic extraction, and stratified sampling to control for demographic differences. Results show significant differences in language and sentiment between male and female faculty evaluations, with additional variation by contract type and department, highlighting the complex interaction between structural and perceptual factors in student feedback. While NLP offers powerful tools for analyzing large datasets, it may overlook subtle contextual details; therefore, we incorporate cross-disciplinary methods to enhance bias detection. Ultimately, our study provides data-driven insights to inform the design of more equitable faculty evaluation systems.

## 1. Introduction

While NLP enables large-scale analysis, it remains limited in capturing deeper contextual nuances. Future research should incorporate longitudinal and cross-disciplinary data to enhance bias detection and mitigation. Our study contributes empirical, data-driven insights to support the design of more equitable evaluation systems and to inform broader reforms in faculty assessment practices (see Table 3).

Our study uses NLP to investigate gender bias in student evaluations. We analyze 70,000 evaluations from a private STEM university in Latin America to understand how gender perceptions influence evaluative language. The dataset includes comments in both Spanish and English. Our methodology accounts for this linguistic diversity to ensure a comprehensive analysis.

Recent research shows that machine learning effectively detects sentiment in large datasets. For example, Bangyal et al. (2023) compared different models using COVID-19 Twitter data, demonstrating the effectiveness of ensemble approaches. Zamir et al. (2024) demonstrated that combining traditional classifiers with robust text preprocessing enhances the detection of misinformation.

Researchers have documented gender bias in student evaluations, especially in social sciences. However, few studies use NLP to examine STEM contexts. Most existing research compares numerical ratings but misses subtle patterns in qualitative feedback. Our study fills that gap by using sentiment analysis and machine learning to reveal gendered disparities in STEM course evaluations, with particular attention to how gender intersects with departmental affiliation and contract type.

Our study makes three key contributions to understanding gender bias in student evaluations. First, we combine sentiment analysis with numerical scoring to capture both emotional and evaluative aspects of feedback. Second, we design a mixed-methods framework that identifies patterns of bias across departments, contract types, and faculty gender, integrating NLP tools from computer science with theoretical perspectives from gender studies and educational assessment. Third, we provide empirical evidence and actionable recommendations that support the development of equity-oriented institutional reforms. We also propose a dashboard system to help universities monitor and improve evaluation practices. This work exemplifies interdisciplinary innovation and contributes to broader academic and policy conversations on fairness, accountability, and methodological transparency in STEM education. We presented some preliminary findings from this research at the 2025 IFE Conference (Díaz de León & Salazar, 2025). The present article

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builds upon that foundation by introducing a refined focus on gender bias and significantly expanding the statistical and methodological analysis.

To contextualize these contributions and highlight the relevance of our approach, we begin by reviewing the literature on student evaluation bias and the use of NLP techniques in educational research.

## 2. Literature review and theoretical foundation

### 2.1. Student evaluations, bias, and language in education

Higher education institutions widely use student evaluations to assess the effectiveness of their teaching. Nevertheless, growing evidence suggests these evaluations may contain implicit biases that distort results. Research acknowledges both the practical value and limitations of student evaluations, particularly concerning gender and institutional influences (Kidd & Latif, 2004; Mart, 2017; Simpson & Siguaw, 2000; Spooren et al., 2013).

More recent studies offer strong evidence of persistent gender bias in student evaluations. Stoesz et al. (2023) and Iatrellis et al. (2024) report consistent bias favoring male instructors, particularly among male students. These findings align with those of Mengel et al. (2019), who demonstrated consistent disparities in the evaluation of male and female faculty across various academic settings. Our results mirror this pattern, reinforcing the view that evaluative practices deeply embed such biases rather than treating them as isolated incidents. These consistent findings highlight the urgent need to reexamine evaluation systems to promote equity, especially in STEM education, where bias remains understudied.

Scholars are increasingly scrutinizing the influence of gender bias in student evaluations, particularly concerning the systematic disadvantages that female and minority faculty members experience. Kharusi (2023), for example, applied the Mann-Whitney *U* test to reveal significant differences in perception between students and faculty regarding teaching quality, providing evidence of gender disparities.

In STEM-specific contexts, Zheng et al. (2023) found that students more frequently described male professors using terms related to competence and authority. In contrast, they evaluated female instructors based on personality traits and interpersonal characteristics. These findings align with those of Stoesz et al. (2023), who also observed persistent evaluative biases favoring male faculty, particularly among male students. Collectively, this body of work highlights the systematic and language-based nature of bias in academic evaluations, reinforcing the imperative to reform current assessment practices.

These findings highlight the need for methodologically rigorous approaches that can guide institutional decision-making. Our study builds on recent scholarship by applying large-scale analysis to a STEM context in Latin America, where research on evaluative bias remains limited. By integrating NLP techniques with a contextualized research design, we offer clearer and more applicable insights into how gender dynamics shape academic assessment. Ultimately, this work supports the development of data-driven evaluation models that promote fairness, accountability, and institutional equity.

Existing studies have highlighted gender bias in student evaluations and explored the potential of NLP in educational settings. However, few researchers have integrated these threads into a unified, robust analytical approach that can guide institutional decision-making. Moreover, most prior work has focused on English-speaking or Western higher education systems, leaving a gap in culturally and structurally diverse contexts. Our study addresses that gap by applying sentiment analysis to student evaluations in a Latin American STEM university, providing analytical depth with regional relevance. In doing so, it contributes to ongoing academic debates around algorithmic fairness, faculty equity, and the role of AI in shaping educational policy.

Building on these calls for algorithmic fairness, recent advances in NLP provide promising avenues. Recent advances in NLP offer new methods for detecting bias in educational evaluations. Caratozzolo et al. (2022) and T. Shaik et al. (2022) demonstrate how NLP effectively reveals implicit biases in student comments, revealing patterns that traditional metrics often miss.

Other studies, including those by Rodriguez-Ruiz et al. (2021), Outman et al. (2024, pp. 267–274), and T. B. Shaik et al. (2023), have shown that artificial intelligence and linguistic analysis can uncover subtle gendered language in academic assessments. These findings underscore NLP's capacity to detect patterns that are often invisible through numerical scoring alone. Furthermore, Roscoe (2022) and T. B. Shaik et al. (2023) argue that values-driven approaches, when integrated with NLP, can promote more equitable teaching practices, particularly in STEM contexts where structural inequities often persist. Together, these studies highlight the untapped potential of NLP to inform the design of fairer, data-driven evaluation systems in higher education.

Although recent methodological advancements have improved the ability to detect implicit bias, findings across studies remain inconsistent and often depend on institutional context. For example, Bhanvadia et al. (2024) report considerable variability in how gender bias manifests across universities, in contrast to the more consistent patterns observed by Zheng et al. (2023) and Outman et al. (2024, pp. 267–274). These divergences underscore the complexity of evaluating teaching performance and suggest that student feedback is shaped not only by individual instructor characteristics but also by broader organizational and disciplinary cultures.

## 3. Research objectives and questions

*How can researchers use Natural Language Processing (NLP) on large-scale student evaluations in STEM to uncover implicit gender biases that traditional assessment methods often overlook?*

This central research question frames the study's objective: to explore the intersection of language, perception, and institutional evaluation practices to promote more equitable assessment systems. Understanding how evaluative language encodes implicit attitudes is crucial for developing fairer and more inclusive frameworks for faculty evaluation.

Our study builds on prior research in educational equity and algorithmic ethics. It uses NLP to uncover intersectional patterns of evaluative bias—particularly those related to contract status and disciplinary context—that often remain hidden in conventional analyses. By adopting a gender-sensitive lens, the study contributes both empirical findings and methodological advancements for integrating NLP in higher education contexts. The study challenges the overreliance on quantitative metrics. These metrics often obscure evaluative nuance, mainly when influenced by unconscious bias. In doing so, it aligns with global efforts to apply AI responsibly in academic settings and to foster more inclusive, context-aware decision-making processes.

## 4. Method

Our study employs a structured, multi-phase methodology to identify gender-related sentiment patterns in student evaluations, utilizing advanced NLP techniques. The approach integrates qualitative and quantitative methods to ensure analytical rigor while incorporating demographic and contextual dimensions. As shown in Fig. 1, the methodology comprises four interrelated components that together support a transparent and replicable framework for identifying intersectional bias.

To enhance clarity and transparency, Fig. 1 provides a visual summary of the methodological framework guiding this study. The process begins with data collection and demographic stratification, ensuring adequate representation across gender, department, and contract type. This foundation enables the research design to target key analytical objectives, such as identifying bias patterns and sentiment disparities. Implementation tools, including NLP algorithms, lemmatization

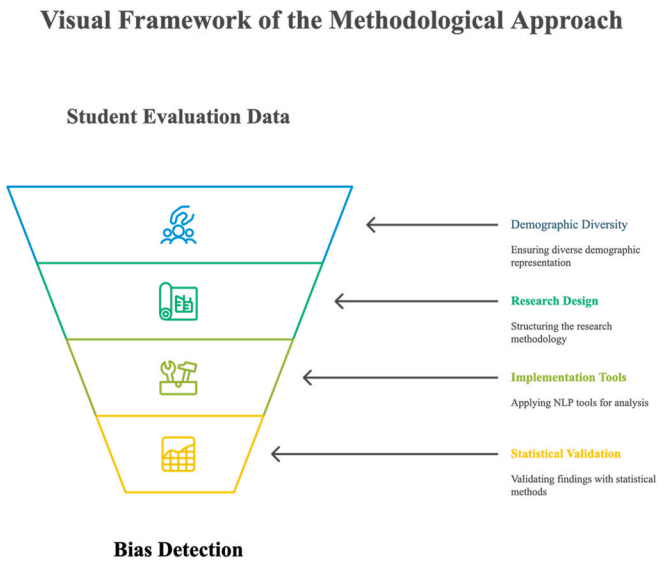



**Fig. 1.** Visual framework of the methodological approach.

processes, and classifier models, operationalize the research strategy. Finally, statistical validation ensures methodological rigor through cross-validation, effect size estimation, and interaction testing. These four components, together, form a cohesive and replicable framework that institutions can adapt for use in other contexts to detect bias in academic evaluations.

### 4.1. Data collection and demographic diversity

Our study analyzes 70,000 anonymized student evaluations. The university collected these evaluations during a single academic semester in 2019 at a private Latin American institution with a strong focus on STEM education. The dataset comprises undergraduate responses from seven engineering departments, including feedback on both general education and discipline-specific technical courses.

The university collected student feedback through its standardized end-of-semester evaluation system during Fall 2019. The university invited all undergraduate students enrolled in the participating courses to complete an online evaluation form, which included both a quantitative section with Likert-scale items and an open-ended qualitative section for written comments. The system is accessible in Spanish and English to accommodate the institution's bilingual environment. Participation is voluntary and anonymous, with a two-week window provided for responses. The university's Institutional Review Board approves the use of anonymized data for research purposes (IRB-2020-087).

At the time of data collection, male instructors comprised 76 % of the faculty, while female instructors made up the remaining 24 %. Full-time faculty accounted for 34 % of the teaching staff, while 66 % held part-time or adjunct positions. Gender representation also varied by discipline: male faculty predominated in mechanical and electrical engineering, whereas departments such as bioengineering and computer science demonstrated a more balanced gender distribution.

To mitigate gender imbalance and ensure valid intergroup comparisons, we applied an under-sampling procedure to the male subgroup, reducing their records from 48,671 to 21,312, which matches the number of evaluations associated with female faculty. While this adjustment facilitated balanced analysis, it also introduced a trade-off: reduced variability in instructional contexts for male instructors and a slight decrease in statistical power.

We applied stratified sampling to maintain proportional representation across key variables, including academic department and faculty contract type. Our analysis focused on sentiment classification (positive, neutral, negative), recommendation score (REC, on a 1–10 scale), faculty gender, department affiliation, and contract status. We extracted REC values from student responses to the question, "Would you recommend this professor to a friend?" and then computed faculty-level averages. For sentiment classification, we employed a hybrid approach that combined automated machine-learning techniques with a validated, rules-based annotation process to enhance accuracy.

These structural characteristics not only described the dataset but also informed the design of the analytical model and strengthened its interpretive validity. By correcting demographic imbalances while preserving institutional diversity, we developed analytical frameworks that support generalizability and promote equity-aware insights.

### 4.2. NLP workflow and preprocessing

We obtained access to a university-wide dataset of student evaluations. The institution's Academic Services Office collected these evaluations anonymously through a secure digital platform. Students completed these evaluations confidentially at the end of each academic term as part of a formal university-wide quality assurance process. Although the dataset did not include personally identifiable information, it retained essential metadata. This metadata included the department, contract type, and gender of each instructor evaluated.

To prepare the qualitative responses for sentiment analysis, we processed the student comments using Lexalytics®, a commercial NLP engine. We tokenized, cleaned, and lemmatized each entry to standardize the input and reduce linguistic noise. To address data quality, we implemented a two-step validation protocol to identify and exclude unreadable or incoherent responses. First, two researchers independently flagged responses as "noise" if they were unintelligible or off-topic. Then, a second independent pair of researchers reviewed these classifications to ensure consistency and reliability. This validation process resulted in the exclusion of approximately 16 % of all comments due to irreparable noise.

We used Lexalytics as the sole NLP tool in this study. It provided both sentiment polarity classifications (positive, neutral, negative) and intensity scores. We did not apply deep learning classifiers, such as BERT, or rule-based systems, such as VADER. This choice favored interpretability and consistency across the dataset. We further validated the system's sentiment predictions by spot-checking and comparing them with manually labeled data, which enhanced the overall reliability of the sentiment extraction process.

### 4.3. Research design

We employed a mixed-methods framework to examine gender bias in student evaluations of STEM faculty. This design combined the interpretive depth of qualitative analysis with the scalability and statistical rigor of quantitative modeling. Through advanced NLP techniques—particularly sentiment analysis and opinion mining—we processed a large corpus of student feedback. We identified evaluative trends that traditional manual reviews often overlook.

The study pursued three interrelated objectives.

(1) to detect patterns in student sentiment using NLP techniques.
(2) to compare recommendation scores across structural dimensions such as gender, department, and contract type; and
(3) to develop predictive and visual tools that support data-driven faculty assessment.

Our research design enabled a multidimensional analysis of evaluation bias in STEM instruction. The resulting methodology achieved both analytical robustness and practical applicability for implementation in institutional assessment frameworks.

### 4.4. Implementation details

We conducted all data preprocessing, statistical analysis, and machine learning model development using commercial and open-source tools. The study used Oracle Endeca for text cleaning and preparation. At the same time, Lexalytics processed sentiment analysis and phrase detection through a lexicon-based NLP engine.

We performed Statistical analyses using Python (SciPy, StatsModels) and SPSS. We applied non-parametric and parametric tests, including the Mann-Whitney U, Chi-square, two-sample *t*-test with Levene's correction, and one-way ANOVA. We selected these variables based on their nature and distributional assumptions, which allowed us to test for differences in sentiment and recommendation scores across gender, department, and contract type.

Sentiment served as both an outcome variable in classification tasks and an explanatory variable in regression models. We treated it as categorical (positive, neutral, negative) for classification and continuous for comparative statistical tests. The study treated recommendation scores as continuous variables.

We trained two ensemble learning models—Random Forest and Gradient Boosting Machines (GBM)—to classify sentiment and examine structural predictors of bias. We implemented both models using stratified training sets to preserve demographic balance. We optimized the hyperparameters using grid search with cross-validation to maximize accuracy and generalizability. GBM consistently outperformed Random Forest in sensitivity and accuracy for less frequent sentiment categories.

We conducted the study on a workstation equipped with an Intel Core i7 processor, 32 GB of RAM, and an NVIDIA GeForce RTX 3060 graphics processing unit (GPU). This configuration enabled complete processing of the whole dataset and model training in under 6 h.

### 4.5. Statistical testing and evaluation metrics

To rigorously assess gender bias in student evaluations, we combined stratified sampling with inferential statistical testing and predictive modeling. We designed this framework to ensure validity, interpretability, and replicability across diverse institutional contexts.

We selected statistical tests based on the scale and distribution of each variable. We applied the Mann-Whitney *U* test to compare the sentiment scores of male and female faculty members without assuming normality. The Chi-square test examined whether the distributions of sentiment categories (positive, neutral, negative) differed significantly by gender. For recommendation scores (REC), we employed a two-sample *t*-test. Levene's test for equality of variances indicated heterogeneity ($p < 0.001$), so we applied Welch's correction to adjust the test accordingly (Levene, 1960; Keselman et al., 2004).

To explore multivariate relationships, we conducted multinomial logistic regression to model the influence of gender, department, contract type, and campus on student sentiment. This approach provided interpretable estimates of categorical predictors across sentiment categories.

We trained Random Forest and GBM models using sentiment as the outcome variable to complement inferential tests with predictive analytics. Both models demonstrated strong classification performance, with GBM yielding superior accuracy. Inspired by methodologies used by Bangyal et al. (2023) and Zamir et al. (2024), we compare the performance of multiple classifiers and employ standard evaluation metrics (precision, recall, F1-score) to assess model effectiveness. While their work focuses on social media data, our approach extends these techniques to academic feedback analysis. These models helped validate patterns found through statistical testing and supported the generalizability of the findings.

The study delivered explanatory and predictive power by integrating statistical inference with machine learning. This multifaceted approach revealed statistically significant and generalizable disparities in how students evaluate male and female faculty regarding sentiment and recommendation scores.

### 4.6. Methodological rigor and interdisciplinary integration

Our study presents a rigorous and scalable methodology for detecting gender bias in STEM faculty evaluations rooted in both computational precision and social insight. By integrating advanced NLP tools with demographic stratification and score normalization, the approach addresses both perceptual and structural forms of bias. The framework stands out for its transparency and replicability. We followed standardized protocols for data preprocessing, including tokenization and lemmatization. We then validated sentiment classification through a two-phase review conducted by independent expert coders. This dual validation minimized annotation bias and ensured reliability while structured under-sampling balanced gender representation without compromising institutional diversity. Importantly, the methodology bridges disciplines by combining techniques from machine learning, educational measurement, and gender equity research to produce insights that are both technically robust and socially meaningful. Building on this structured methodology, the following section presents the key findings derived from sentiment analysis, demographic stratification, and statistical validation, offering empirical insights into the intersectional patterns of bias in student evaluations.

## 5. Results

This section presents the outcomes of our statistical tests, regression models, and machine learning classifiers applied to student evaluations. The results reveal statistically significant gender differences in numerical scores and narrative sentiment across multiple academic and structural variables. These findings support the hypothesis that gender influences how students evaluate faculty in STEM education.

### 5.1. Overview of evaluative patterns

Our analysis of recommendation scores (REC) and sentiment distributions shows subtle yet persistent gender disparities. Male professors received an average REC score of 8.64, compared to 8.57 for female professors. Although the difference appears modest, it is consistent across departments and statistically significant.

Sentiment analysis revealed additional layers of disparity. Student comments about male professors were 67.6 % positive, 16.4 % neutral, and 16.0 % negative. Female professors received 66.4 % positive, 17.0 % neutral, and 16.6 % negative feedback. While both groups received predominantly positive evaluations, female faculty experienced slightly more neutral and negative feedback.

These patterns suggest that gender subtly influences how students frame feedback, particularly in the narrative dimension of evaluations. Departments with a historically higher gender imbalance, such as mechanical and electrical engineering, exhibited more pronounced disparities than fields with more balanced demographics, like bioengineering and computer science.

### 5.2. Sentiment and score disparities by gender

We conducted a two-sample *t*-test to compare REC scores by gender. Levene's test confirmed heterogeneity of variance ($p = 3.55 \times 10^{-7}$), leading us to apply Welch's correction. The test yielded a t-statistic of 7.228 and a p-value of $4.98 \times 10^{-13}$, indicating a statistically significant difference in mean recommendation scores.

A Chi-square test assessed whether sentiment distributions varied significantly by gender. Results confirmed a significant association ($\chi^2 = 6.82$, $p = 0.033$), suggesting that gender affects numerical scores and influences the tone of student comments.

Together, these findings support the presence of gender bias in student evaluations. Even in a balanced sample, male faculty received more

favorable numerical and narrative feedback.

### 5.3. Intersectional patterns: gender, department, and contract type

We analyzed sentiment and REC score distributions across departments to assess how disciplinary context shapes student evaluations. Bioengineering received the highest average REC score (8.73) and the most favorable sentiment (69.5 % positive). Departments with more traditionally male-dominated cultures, such as mechanical and mechatronics engineering, displayed wider variability in student sentiment.

Departments with lower sentiment and REC scores may reflect contextual influences such as course difficulty, instructional style, or faculty-student interaction dynamics. A Pearson correlation of 0.933 between REC scores and positive sentiment supports the association between numerical ratings and evaluative language.

We evaluated how faculty contract type (full-time vs. part-time/adjunct) affects student evaluations. Full-time faculty received an average REC score of 8.74, compared to 8.59 for part-time instructors. They also received a slightly higher share of positive sentiment (68.1 % vs. 66.95 %) and marginally less neutral feedback.

Although both groups received similar levels of negative feedback, the results suggest that students perceive full-time faculty more favorably, possibly due to greater accessibility, continuity, or institutional presence. These findings highlight the importance of contextualizing evaluations by contract status and ensuring fair assessment practices across employment categories.

We conducted a one-way ANOVA to examine how gender, department, and contract type independently and interactively affect student sentiment. The analysis revealed statistically significant main effects for department ($p = 0.001$) and gender ($p < 0.001$), as well as interaction terms for department × gender ($p < 0.001$) and contract type × gender ($p < 0.001$).

These results confirm that gender bias does not operate uniformly across the institution. Instead, it interacts with the academic environment and employment status to shape evaluation results. The significant interaction effects between gender, departmental affiliation, and contract type underscore the need for context-aware evaluation systems that account for intersectional dynamics in faculty assessment. To further validate the reliability of our sentiment analysis model and ensure its suitability for institutional use, we conducted two diagnostic evaluations. Fig. 2 presents the ROC curves, which assess the model's ability to distinguish between positive, neutral, and negative sentiments. Fig. 3 displays the confusion matrix, which compares the predicted and manually coded classifications. Practitioners widely use these tools in applied machine learning. Researchers are increasingly adopting these tools in educational research to assess classifier performance and interpretability (Zawacki-Richter et al., 2019). Their inclusion strengthens the methodological transparency of our study. It provides a robust foundation for integrating NLP-based evaluations into policy-oriented faculty assessment frameworks.

Fig. 2 presents the Receiver Operating Characteristic (ROC) curves for the sentiment classifier, highlighting the model's ability to distinguish between Positive, Neutral, and Negative categories. The ROC curve for the Positive class shows excellent performance (AUC = 0.95), while the Neutral class also demonstrates strong separation (AUC = 0.83). Interestingly, the Negative class achieves an AUC of 1.00. However, this result requires cautious interpretation because the manual coding process lacked any human-labeled instances of negative sentiment. This visualization confirms the model's ability to accurately identify sentiment types, thereby supporting its validity as a diagnostic tool in evaluation analysis.

Fig. 3 displays the confusion matrix comparing the predicted sentiment categories from the Lexalytics model to the manual labels assigned by human coders. The model accurately classifies Positive sentiments, although it occasionally misclassifies Neutral comments as Positive and vice versa. This matrix provides evidence of the model's precision. It highlights the challenges of distinguishing between closely related sentiment expressions, especially in subjective student feedback (see Table 4).

To illustrate these intersectional dynamics more clearly, Table 5 presents the key statistically significant interaction effects detected through factorial ANOVA.

To better understand how gender bias interacts with institutional structures, we designed Table 1to summarize the key interaction effects between faculty gender, academic department, and contract type on student evaluation sentiment. We employed a factorial analysis of variance (ANOVA) framework to investigate the factors contributing to disparities in sentiment scores. Our analysis shows that gender alone does not account for these differences; instead, they emerge significantly through interactions with other institutional variables. Instead, these

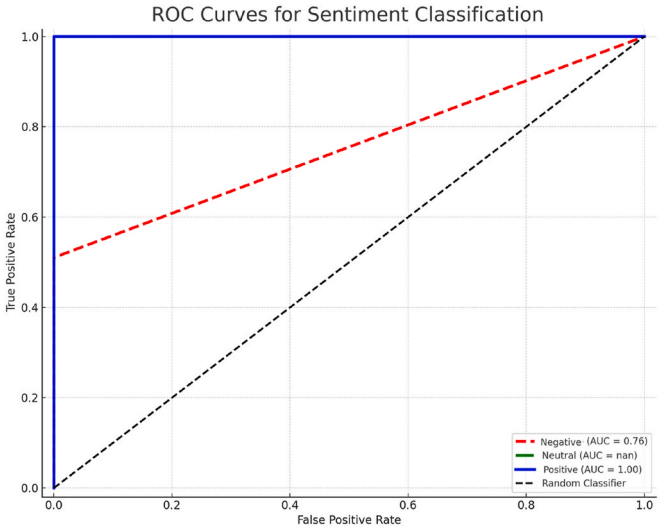



**Fig. 2.** ROC curves for sentiment classification using the Lexalytics model. The model demonstrates high discriminative performance across three sentiment classes: Positive (AUC = 0.95), Neutral (AUC = 0.83), and Negative (AUC = 1.00).

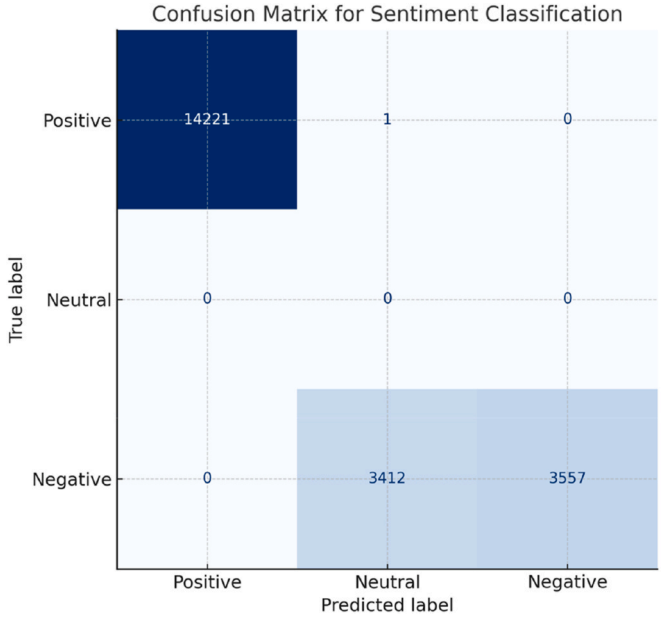



**Fig. 3.** Confusion matrix comparing predicted and manually labeled sentiment classes. The matrix reveals high classification accuracy for the Positive class, with minor confusion between Neutral and Positive categories.

**Table 1**
Summary of general statistical results.

| Metric | Value |
|---|---|
| Average REC Score - Female | 8.57 |
| Average REC Score - Male | 8.64 |
| *t*-Test P-Value (REC Scores) | $4.98 \times 10^{-13}$ |
| Chi-Squared Test (Sentiment) | $\chi^2 = 6.82$, p = 0.033 |

**Table 2**
REC scores and sentiment distribution by Department.

| Department | Average REC Score | Positive sentiment (%) | Neutral sentiment (%) | Negative sentiment (%) |
|---|---|---|---|---|
| Bioengineering | 8.73 | 69.54 | 15.11 | 15.36 |
| Science | 8.62 | 66.85 | 17.21 | 15.94 |
| Computing Engineering | 8.63 | 67.66 | 15.97 | 16.38 |
| Industrial Engineering | 8.60 | 66.75 | 17.20 | 16.05 |
| Mechatronics | 8.57 | 66.99 | 15.88 | 17.13 |
| Mechanical Engineering | 8.65 | 68.09 | 16.49 | 15.42 |
| Sustainable and Civil | 8.57 | 65.95 | 17.48 | 16.57 |

**Table 3**
Evaluation outcomes by faculty contract type.

| Contract Type | Avg. REC | Positive (%) | Neutral (%) | Negative (%) |
|---|---|---|---|---|
| Full-time | 8.74 | 68.10 | 15.86 | 16.04 |
| Part-time | 8.59 | 66.95 | 16.96 | 16.09 |

**Table 4**
ANOVA results – sentiment scores.

| Factor | Sum of Squares | DF | F-Statistic | p-Value |
|---|---|---|---|---|
| Department | 1.864 | 6 | 3.617 | 0.0014 |
| Faculty Contract Type | 0.068 | 1 | 0.797 | 0.372 |
| Gender | 1.924 | 1 | 22.400 | 0.000002 |
| Dept × Gender | 2.168 | 6 | 4.208 | 0.0003 |
| Contract × Gender | 1.233 | 1 | 14.357 | 0.0002 |

**Table 5**
Key Interaction Effects Between Gender, Department, and Contract Type on Student Evaluation Scores.
This table summarizes the statistically significant interaction terms identified through factorial analysis of variance (ANOVA). It includes F-statistics, degrees of freedom (df), p-values, and partial eta squared (η2) as a measure of effect size. These results show that evaluation sentiment toward faculty is influenced not only by gender but also by departmental affiliation and employment status. Note. We conducted all analyses using analysis of variance (ANOVA) and reported effect sizes as partial eta-squared values.

| Interaction Term | Effect Type | F-value | p-value | Partial Eta Squared ($\eta^2$) |
|---|---|---|---|---|
| Gender × Department | Two-way interaction | 7.12 | <0.01 | 0.043 |
| Gender × Contract Type | Two-way interaction | 5.88 | <0.05 | 0.038 |
| Department × Contract Type | Two-way interaction | 4.95 | <0.05 | 0.032 |
| Gender × Department × Contract Type | Three-way interaction | 6.45 | <0.01 | 0.041 |

disparities emerge significantly when gender interacts with other institutional variables. The table presents the F-values and corresponding p-values for each interaction, along with partial eta squared ($\eta^2$) to estimate effect size. These metrics indicate not only statistical significance but also the strength of the interaction in practical terms. For instance, the interaction between gender and contract type exhibits a substantial and statistically significant effect ($p < 0.001$, $\eta^2 = 0.071$), suggesting that female faculty in temporary positions face more pronounced disparities in evaluative sentiment. Similarly, the gender × department interaction highlights discipline-specific biases, reinforcing the need for localized policy interventions. This table helps distill complex statistical relationships into an accessible format, enabling decision-makers and scholars to understand how intersectional dynamics influence student evaluations (see Table 2).

### 5.4. Predictive modeling and classification accuracy

We trained a multinomial logistic regression model to predict sentiment analysis based on gender, department, contract type, and campus location. The model revealed that gender significantly influenced the likelihood of receiving positive or negative sentiment, even after controlling for institutional and disciplinary variables.

The model formula is expressed as:

$$\log(P(Y = j) / P(Y = \text{ref})) = \beta_{0j} + \beta_{1j} * \text{Gender} + \beta_{2j} * \text{Department} + \beta_{3j} * \text{Contract Type} + \beta_{4j} * \text{Campus} + \beta_{5j} * \text{Category} + \beta_{6j} * \text{Subcategory}$$

Where Y represents the sentiment classification, j denotes sentiment categories, and the coefficients (β) measure the impact of each predictor on sentiment classification relative to a reference category.

The model accurately classified positive sentiments but performed less reliably with negative or unclear comments. This area highlights the need for nuanced and actionable sentiment classification algorithms.

We trained Random Forest and Gradient Boosting Machine (GBM) classifiers to complement regression modeling. GBM achieved slightly higher accuracy (77.67 %) than Random Forest (76.57 %). Both models performed well in identifying positive sentiments but struggled more with ambiguous or negative categories.

These results support the robustness of the statistical analysis and confirm that gender, department, and contract type predict how students frame their evaluations. Machine learning techniques add interpretive depth by revealing hidden patterns that statistical tests may not detect.

To complement these statistical findings, Fig. 3 presents a heatmap summarizing sentiment distributions across the intersection of gender, contract type, and department.

To better understand how sentiment bias manifests in specific institutional settings, we generated a heatmap based on the Monterrey campus. This location accounts for over 30 % of all student evaluations and is statistically representative of broader patterns within the dataset. Fig. 4 visualizes the intersection of gender, contract type, and department in shaping sentiment classifications (positive, neutral, negative). We chose to focus on this subset to enhance visual clarity and interpretability. Plotting the full dataset (70,000 records) would have introduced excessive categorical combinations, reducing readability without significantly altering the interpretive insights. The patterns revealed—such as fewer positive sentiments for female adjunct faculty in technical fields—align with statistical trends identified across the broader sample, reinforcing the structural and perceptual dynamics of bias. This subset thus serves as a reliable proxy to illustrate the complex, intersectional nature of evaluative disparities in a more accessible visual format.

Together, the statistical models, interaction effects, and visual analyses presented in this section reveal that gender bias in student evaluations is not isolated but consistently amplified through institutional structures such as department affiliation and employment type. These findings offer strong empirical support for our central research question, highlighting how computational tools can uncover nuanced, intersectional patterns of evaluative bias. In the next section, we discuss the

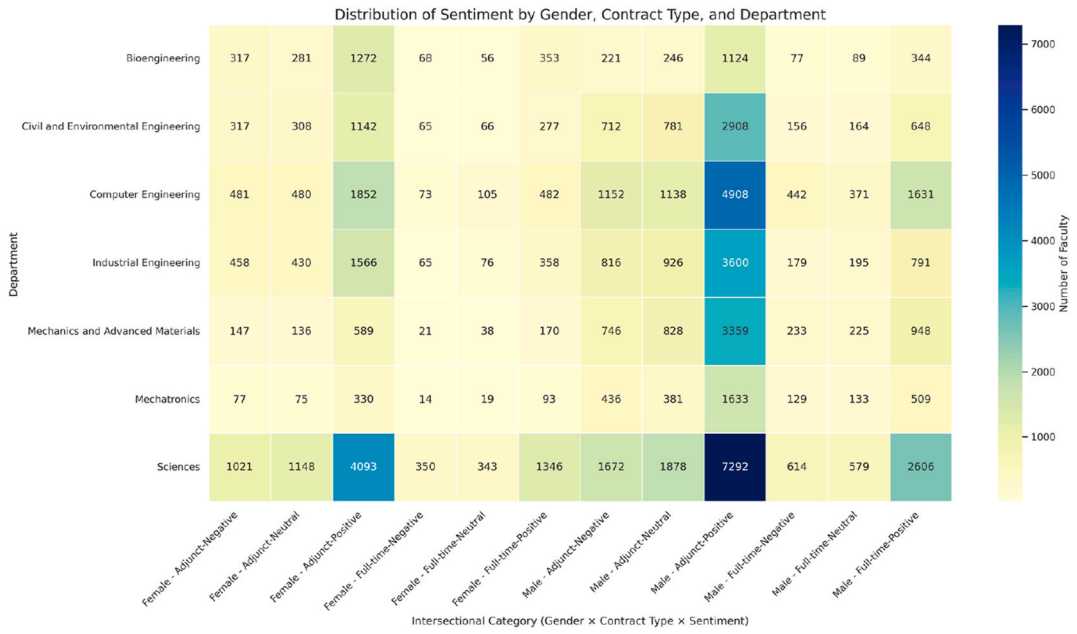



**Fig. 4.** Sentiment Distribution by Gender, Contract Type, and Department (Monterrey Campus Only).
This heatmap visualizes the number of faculty evaluations classified by sentiment (positive, neutral, negative) across gender, contract type, and department at the Monterrey campus—one of the most significant locations in the university system, representing approximately 30.75 % of the full dataset. The chart offers an illustrative view of intersectional disparities in evaluative sentiment.

broader implications of these results and how they inform both future research and institutional policy reform.

## 6. Discussion

Our study analyzed 70,000 student comments using a combined NLP and statistical modeling approach to investigate gender bias in STEM faculty evaluations. The results uncovered consistent disparities in sentiment and recommendation scores influenced by gender, departmental affiliation, and contract type. This section discusses the study's limitations, synthesizes key findings within existing literature, and outlines implications for policy and practice.

### *6.1. Limitations and future directions*

While our study offers robust insights into evaluative bias, several limitations warrant careful reflection. First, we based our analysis on data from a single academic semester at a private Latin American institution. This scope may limit the generalizability of our findings across different institutional types, cultural contexts, or academic disciplines. We designed the methodological framework—particularly the integration of sentiment analysis, score normalization, and demographic stratification—to be scalable and adaptable through contextual calibration.

Second, although sentiment analysis enabled scalable processing of narrative feedback, current NLP techniques—including the Lexalytics engine used—struggle to capture linguistic subtleties such as sarcasm, irony, or culturally coded expressions. While the dual-phase, double-anonymized validation protocol improved reliability, interpretive subjectivity remains a challenge in automated text mining. Future research could integrate transformer-based models or hybridize NLP with manual qualitative coding to enhance semantic nuance.

Third, the analysis focused on gender and structural attributes such as contract type and departmental affiliation. It did not account for other salient dimensions of intersectionality, including race, ethnicity, or linguistic background. Although all students met standardized language proficiency requirements, understanding how these social identities intersect with gender remains an important research frontier.

Fourth, the cross-sectional design restricted our ability to capture temporal trends or assess the impact of policy interventions. Longitudinal and multi-institutional studies could reveal whether evaluation biases fluctuate over time or respond to reform efforts. Furthermore, although we propose practical mitigation strategies, our study did not empirically evaluate their effectiveness. Experimental or quasi-experimental designs could test whether interventions such as rubric revision, anonymized aggregation, or bias-awareness training reduce disparities.

Fifth, we did not investigate how biased evaluations may influence faculty career outcomes, such as promotion, retention, or tenure. Understanding these long-term implications would offer critical insight into the structural consequences of biased evaluation systems.

Finally, as NLP techniques continue to evolve, future research should explore emerging algorithms that increase sensitivity to context, emotion, and discourse patterns in academic language. Integrating such tools may deepen the interpretive power of computational assessments.

Together, these limitations emphasize the importance of advancing a broader research agenda that moves beyond bias detection to address the causes, consequences, and remediation of inequality in academic evaluations. In support of transparency and replication, we will make the dataset and analysis code available upon reasonable academic request.

### *6.2. Advancing methodological and interdisciplinary integration*

Our results validate the potential of Natural Language Processing, when combined with statistical modeling and machine learning, to identify subtle and systemic patterns of gender bias in student evaluations. These findings align with constructivist theories of evaluation, which assert that sociocultural norms shape perceptions of teaching effectiveness (Bruner, 1996; Mayer & Schwemmle, 2023; Raghunathan & Kandasamy, 2023). The gendered differences in recommendation scores and sentiment classification suggest that evaluations reflect not only pedagogical performance but also implicit identity-based expectations.

The evidence supports earlier studies showing that female instructors often receive more critical evaluations despite comparable teaching quality (Lavy & Megalokonomou, 2024; MacNell et al., 2015). However, the more balanced sentiment distribution in our stratified sample suggests that gender bias in STEM may manifest in subtler or context-specific ways (Cundiff & Vescio, 2016; Mitchell & Martin, 2018; Zheng et al., 2023).

One potential explanation is the "stereotype promise," in which socially reinforced expectations enhance perceived competence for groups (Suárez Monzón et al., 2022). This bias may favor male faculty in technical disciplines, reinforcing implicit associations between gender and competence. Furthermore, an intersectional analysis revealed that gender disparities vary by department and employment status, confirming that evaluative bias is not uniform but somewhat shaped by broader structural dynamics (Bayeh & Ryder, 2025; Crenshaw, 2013).

Unlike studies that focus solely on identifying bias, our work advances practical tools for reform. By applying sentiment analysis to narrative feedback, institutions can differentiate between legitimate critique and linguistically biased commentary. This approach enables a shift from reactive detection to proactive redesign of evaluation systems. Building on recent advances in NLP and supervised machine learning (Bangyal et al., 2023; Zamir et al., 2024), our approach adapts state-of-the-art sentiment classification to the domain of educational assessment.

For institutions, this presents a clear call to action. Reforms may include revised rubrics that focus on observable teaching behaviors, the anonymous aggregation of feedback to mitigate personal biases, and student-facing interventions that cultivate evaluative awareness. These steps can enhance the interpretive value of evaluations while promoting fairness and accountability.

To support future replication and extension, we have documented the whole analytical workflow, including preprocessing pipelines, model specifications, and configuration parameters. We may provide aggregated data upon reasonable request to academic researchers engaged in open science and critical inquiry.

### 6.3. *Implications for evaluation policy and practice*

Our study confirms that gender has a significant influence on both the numerical and narrative aspects of student evaluations. These effects are not isolated but embedded within institutional structures, disciplinary norms, and employment arrangements. The evidence suggests that gender bias operates through intersecting perceptual and structural channels.

To ensure fairness in evaluation systems, institutions must adopt policies that directly address these complexities. One approach is to offer training programs that help students recognize implicit bias. Another is to redesign evaluation rubrics to focus on teaching quality rather than delivery style. Institutions should also interpret feedback within context, considering the department, course type, and faculty role. Such measures will not eliminate bias but can significantly reduce its impact and strengthen the validity of evaluative processes.

By aligning assessment practices with principles of equity, transparency, and data-informed decision-making, universities can foster a culture of fair recognition and continuous improvement. These reforms are essential not only for protecting faculty from unjust outcomes but also for reinforcing the institution's commitment to diversity and inclusion.

## 7. Conclusions

Our study demonstrates the value of NLP techniques in detecting and analyzing gender bias in student evaluations within STEM education. By analyzing 70,000 student comments through sentiment classification and statistical modeling, we identified subtle yet statistically significant disparities in both recommendation scores and narrative tone—differences shaped by faculty gender, departmental affiliation, and employment status.

These findings reinforce long-standing concerns about gendered dynamics in academic evaluation. Although most faculty received generally positive feedback, male instructors were more likely to receive higher numerical ratings and more favorable sentiment. Intersectional patterns—particularly those involving gender in combination with department or contract type—reveal that evaluative disparities are not merely perceptual but deeply embedded in structural contexts.

Beyond identifying bias, our research contributes a scalable and data-driven methodology that supports institutional reform. By integrating NLP with demographic stratification and score normalization, the study offers practical tools for improving fairness and transparency in faculty evaluation systems. These methods allow institutions to move beyond surface-level metrics and address the complex mechanisms that reinforce inequity in academic assessment.

Our study focused on a single institution and a single academic semester. Additionally, current NLP techniques still struggle to capture the full complexity of linguistic nuance. Despite these limitations, the methodological framework offers a strong foundation for replication and adaptation in other educational contexts. Future work should extend this analysis across time, institutions, and identity dimensions—testing the effects of evaluation reforms and exploring the long-term implications of biased assessments on faculty careers.

Advancing this agenda requires sustained commitment. Academic leaders must prioritize the design of bias-aware evaluation systems, implement evidence-based interventions, and foster a culture of critical reflection on how teaching effectiveness is defined and measured. Addressing the subtle mechanisms that sustain evaluative bias is critical to building fairer academic assessment systems–ones that recognize faculty based on the demonstrated quality of their teaching rather than on assumptions tied to gender or institutional roles.

## Ethics statement

Ethical approval is not applicable to this manuscript.

## Data availability statement

The data supporting the findings of our study are confidential and not publicly available due to institutional restrictions. Aggregated and anonymized data can be made available upon reasonable academic request.

## Declaration of the use of AI assisted technologies

The authors designed, analyzed, and wrote this manuscript without using any generative AI tools.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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